11-30 第2篇

阅读理解（说明文） Why Do Onions Make Us Cry?

① On Saturday afternoon, while helping Mum prepare dinner, I cut a medium onion in half. Within about ten seconds, tiny silver **tears** appeared at the corners of both my eyes. My nose tingled. I laughed, a little embarrassed. Mum laughed too. "This is one of the small mysteries of the kitchen, Ming. **An onion can make a grown man cry even when he is in a very good mood.**" Behind the joke, however, sits a neat piece of biology and chemistry worth understanding. Why exactly does a fresh-cut onion make our eyes water, and what can we do about it?

② First, the onion itself. Onions belong to the same family as garlic and leek. Deep inside an onion's cells, even when it sits quietly on the shelf, there are two separate things: (1) small **sulfur-containing molecules** called **S-oxides**, and (2) a small enzyme called **alliinase**. Both live in the onion, but in **different compartments inside the cell**. As long as the cell is whole, they never meet — **no smell, no tears, no trouble**. A whole onion is almost shy.

③ Now, we pick up the knife. When the blade breaks open the onion cells, the compartments burst. The S-oxides and the enzyme alliinase suddenly meet. Within a small fraction of a second, they produce a **new, unstable chemical** called **syn-propanethial-S-oxide** — a complicated name for a very simple thing: **a small, flying molecule of irritating gas**, light enough to rise from the chopping board into the air and reach your eyes in a breath.

④ What does this gas do when it reaches the eye? The surface of the eye is always covered by a **thin layer of water** (our tear film). The small irritant gas **dissolves into this water** and reacts with it, forming a weak form of **sulfuric acid** — a very small amount, far less than a laboratory acid, but enough to hurt the sensitive nerves on the surface of the eye. The nerves send a quick signal to the tear glands: **"Flush!"** The glands respond immediately with extra tears, trying to **wash the irritant away**. That extra water is the "onion crying" you see.

⑤ There are **five easy kitchen tricks** to reduce this reaction, each based on the chemistry above. **(1) Cool the onion.** Putting the onion in the fridge for 15—20 minutes before cutting slows down the enzyme. A cold alliinase is a slow alliinase. **(2) Cut under running cold water** or near a bowl of water. The gas dissolves in water **before** it reaches your eyes. **(3) Use a sharp knife.** A sharp blade cuts through each cell cleanly; a dull blade tears more cells, making more gas. **(4) Cut the root last.** The highest concentration of S-oxides is near the root (the small dry end). Removing the root last

means less gas for less time. **(5) Turn on a small fan** or open a window. Moving air carries the gas away from your face.

⑥ **Do onions cry to protect themselves?** Yes — this is the most beautiful part of the story. In the wild, insects and small animals that bite an onion bulb suddenly get a strong, irritating gas in their mouth and eyes. Most of them let go. Over millions of years, onions that happened to make more irritating gas survived better, passed on their genes, and slowly, carefully, built this tiny chemical defence system. So the small tear in our eye is, in fact, a distant echo of **an old plant's long project to keep itself alive**. When you eat an onion, you are eating a plant that, very politely, has already tried to warn you.

⑦ A few more interesting facts. **(a) Sweet onions** (like the sweet yellow kind from Vidalia, Georgia, in the US) contain **less sulfur** and make fewer tears — this is not a different species, just the same onion grown on low-sulfur soil. **(b) Cooked onions do not make us cry**; heat destroys both the enzyme and the unstable gas. **(c) Garlic does not cut into tears** — garlic makes a stronger-smelling chemical (allicin) but one that is **less volatile**, meaning it does not easily rise into the air toward your eyes. **(d)** Some scientists at the University of Sydney in 2008 actually **made a "no-tears onion"** by switching off one gene; it tasted almost the same but carried far less S-oxide gas. The tear-free onion exists; it just has not reached every kitchen yet.

⑧ A small safety note. If an onion's gas ever reaches your eyes strongly — after peeling a big bag of onions, for example — the best response is **not to rub your eyes**. Rubbing pushes more irritant into the surface and makes the tears worse. Simply **walk to a window, take slow breaths, and rinse your eyes with clean running water** for thirty seconds. The tear glands will soon do the rest of the work themselves.

⑨ So the next time you are standing at a chopping board, holding half an onion, and the small silver tears appear at the corners of your eyes, remember: you are caught in a tiny, very old conversation between a plant and an animal. The plant, for millions of years, has been quietly saying *"please do not eat me"* in the only way it knows; your eye, bravely covered in a thin layer of water, is simply doing its honest job of washing a small amount of unwanted chemistry away. A fresh onion does not really make us sad. **It makes us briefly, accidentally, part of a story much older than our kitchen.** Smile. Keep cutting. Then have dinner.

注释： enzyme 酶；sulfur 硫；sulfuric acid 硫酸；irritate 刺激；volatile 易挥发的

根据短文内容，选择最佳答案。

1. Why does a WHOLE onion NOT make us cry?

A. Because whole onions have no chemicals.
B. Because the S-oxides and the enzyme alliinase live in different compartments inside

the cell; as long as the cell is whole, they never meet, so there is no smell, no tears, no trouble.
C. Because whole onions are not real onions.
D. Because the tear is caused by light only.

2. What exactly does the "irritant gas" from a cut onion do when it reaches the eye?

A. It simply smells bad.
B. The surface of the eye is covered by a thin layer of tears (water); the gas dissolves in this water and reacts to form a very small amount of weak sulfuric acid, which irritates the sensitive eye nerves; they signal the tear glands to "flush," producing extra tears to wash the irritant away.
C. It freezes the eye.
D. It makes the eye warm.

3. Which of the following is one of the five kitchen tricks to reduce crying?

A. Warm the onion in the oven.
B. Cut the onion with a dull knife.
C. Cool the onion in the fridge for 15—20 minutes before cutting, because a cold enzyme is a slow enzyme.
D. Hold the onion very close to your face.

4. Why, from an evolutionary point of view, do onions make an irritant gas?

A. To attract insects.
B. As a tiny chemical defence — over millions of years, onions that made more irritating gas when bitten survived better (insects and small animals let go), passed on their genes, and built this small defence system; our tears today are a distant echo of the plant's long project to stay alive.
C. To decorate our dinner.
D. To make humans stronger.

5. If onion gas reaches your eyes strongly, what should you do?

A. Rub your eyes hard until the tears stop.
B. Do NOT rub your eyes (rubbing pushes more irritant into the surface); walk to a window, take slow breaths, and rinse with clean running water for thirty seconds; the tear glands will do the rest.
C. Close your eyes and sleep for an hour.
D. Wash your eyes with soap.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段"两种物质分室存放"
【解析】直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第④段作用机制
【解析】溶于泪 + 生成弱酸 + 冲洗反射直引。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第⑤段五条方法
【解析】冷藏 15—20 分钟直引。A、B、D 均错。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段进化解释
【解析】"化学防御"主旨直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑧段安全提示
【解析】"不揉眼 + 清水冲洗 30 秒"直引。A、C、D 均错。故选 B。